

Performance Testing Report

Personalized Networking Assistant | Load & Stress Test Documentation

Date	30 June 2026
Team ID	XXXXXX
Project Name	Personalized Networking Assistant
Maximum Marks	5 Marks

Performance testing evaluates how the system behaves under expected and peak load conditions. This report documents the testing approach, tools used, and results obtained for the **/generate** API endpoint of the Personalized Networking Assistant.

Step 1 — Testing Overview

Field	Details
Testing Tool Used	Postman
Type of Testing	Load Testing, Stress Testing
Target Module / API	/generate API Endpoint
Test Environment	Local Flask Server (localhost:5050)
Test Date	15 March 2026

Step 2 — Test Scenarios

S.No	Test Scenario / Description	Virtual Users	Duration (sec)	Expected Outcome
1	Generate captions for Instagram (Casual Tone)	10	60	Captions generated successfully
2	Generate captions for LinkedIn (Professional Tone)	20	60	System handles concurrent requests without failure
3	Multiple users generating captions simultaneously	30	120	Response time remains acceptable
4	Continuous requests to /generate endpoint	50	180	API remains stable with minimal errors

Step 3 — Performance Test Results

S.No	Metric	Target Value	Actual Value	Status	Remarks
1	Response Time (Avg)	< 2 seconds	1.6 seconds	Pass	Within acceptable range
2	Response Time (Max)	< 5 seconds	4.2 seconds	Pass	Delay due to AI inference
3	Throughput (Req/sec)	10 Req/sec	12 Req/sec	Pass	Handled expected load
4	Error Rate	< 1%	0.4%	Pass	Few timeout errors observed
5	CPU Utilization	< 80%	65%	Pass	CPU usage remained stable
6	Memory Utilization	< 80%	58%	Pass	Memory usage within limits

Step 4 — Observations & Analysis

- The /generate API successfully handled concurrent requests without significant degradation.
- Average response time remained below the target threshold.
- Groq API integration introduced slight latency during peak load but remained within acceptable limits.
- No major system crashes or failures were observed during testing.
- CPU and memory consumption stayed within safe operational limits.
- Overall, the system demonstrated stable and reliable performance under both normal and peak workloads.

Step 5 — Screenshots / Evidence

Attach screenshots of the performance testing tool results below (e.g., Postman Collection Runner summary, response-time graphs, server logs):

[Insert testing tool screenshots here]